

Timeouts and retries

2026-08-14. Numbers are the shipped [config/tuning.yaml](#).

Four independent layers can each abandon work, and each has its own retry. They nest, and their retries **multiply** — which is the part that is easy to get wrong when tuning any single number.

(Diagrams are plain ASCII on purpose: box-drawing glyphs are outside Courier's WinAnsi encoding and render substituted or blank in the PDF.)

The layers

```
+== TURN ===== turn_wall_clock_s = 360s ==+
| asyncio.wait_for on the whole turn. On expiry the coroutine is CANCELLED, |
| so every agent still in flight dies with it -- they report `cancelled`, |
| not a timeout of their own.          no retry: the turn ends |
|
| +- PHASE ----- one fence per kind of agent -----+ |
| | screen      30s      orch_plan / reviewer    25s      | |
| | specialist  240s     report_agent            150s     | |
| | distiller   120s     distiller_drain          60s      | |
| |
| | Expiry = that agent failed. The turn continues, degraded. | |
| | orch_plan is the only one that retries (round-1 watchdog). | |
| |
| | +- AGENT ----- _MAX_SPECIALIST_ATTEMPTS = 2 -----+ | | | | | |
| | | Re-runs the whole agentic loop when the ANSWER is wrong: | | |
| | |   ungrounded_retry   built on a failed tool call      | | |
| | |   absence_reread     denies what the tools returned   | | |
| | |   max_turns_retry    blew its turn budget             | | |
| | |
| | | +- CALL ----- one HTTP request to the model -----+ | | |
| | | | stall fence  40s -> abandon, re-issue ONCE          | | | |
| | | | full budget  60s -> TimeoutError + safechain_retry_ | | | |
| | | |                                     stalled (the falsifying datum) | | | |
| | | | HTTP 401      -> refresh token, retry IN PLACE      | | | |
| | | | 403 / 400     -> FirewallRejection, escalates to the | | | |
| | | |                                     guidance loop below | | | |
| | | |
| | | +-----+
| | +-----+
| +-----+
+=====+

+- REJECTION (wraps the call) ----- firewall.max_retries = 2 -----+
| 403 compliance block / 400 bad request are DETERMINISTIC, so an |
| identical retry is guaranteed to fail the same way. This loop |
| retries with a CHANGED input instead -- `_inject_guidance` appends |
| firewall guidance telling the model what to remove. 401 is the |
| opposite case (a stale token, transient) and is retried in place. |
+-----+
```

Also in the path, but not a fence: the firewall semaphores
(specialist 8, orchestrator 4).

The call layer, in detail

The layer added on 2026-08-13, and the only one that treats a slow call as *suspect* rather than as work in progress.

```
issue request
|
```

```

+-- returns < 40s -----> ANSWER (p99 is 7.5s)
|
+-- still running at 40s
    | the call is not slow, it is WEDGED
    | log: safechain_call_stalled
    v
    abandon it (ainvoke is genuinely cancellable, so the
                request aborts -- no orphaned thread)
    |
    issue a FRESH request, full 60s budget
    |
    +-- returns -----> ANSWER (~45s total)
    |
    +-- stalls too -----> FAILS at 100s
        | logs safechain_retry_stalled
    +-- exactly one extra attempt; never a loop

```

The budget bets that the retry ESCAPES. The alternative was a budget above the 126-131s plateau so a stalled retry could ride it out — safer, but it makes every persistent wedge cost ~170s. The bet is supported by concurrent divergence: `distiller.bureau` returned in 5.8s while `distiller.modeling` hung 120s in the same pool at the same moment, so the wedge belongs to the REQUEST and a fresh one should not inherit it. Given that, the budget only has to cover a healthy call — p99 7.5s in dev, 2.1-5.6s in prod, so 60s leaves ~8x margin.

	before	after
retry escapes the stall	~130s	~45s
retry stalls too	~130s	fails at 100s, logged
call is healthy	unchanged	unchanged, issued once

The bet is falsifiable. `safechain_retry_stalled` fires only when a RE-ISSUED request also timed out. Zero occurrences means the budget can drop further; anything non-trivial means the wedge survives re-issue and the budget must go back above the plateau.

Why 40s and not tighter. Per-call latency is p99 7.5s over 88 dev calls and 2.1-5.6s in prod, so 40 is ~5x the observed worst case: it cannot abandon a healthy call, and it only governs how long a STALL burns before the retry.

`SAFECHAIN_STALL_RETRY_S=0` disables it. The first attempt is clamped to `min(stall_fence, full_budget)`, so lowering `SAFECHAIN_CALL_TIMEOUT_S` really does lower it — unclamped, the fence silently outlived a tighter setting.

Why this layer exists

Measured in the private env: safechain calls do not run slow, they **stall**. In one turn `distiller.bureau` returned in 5.8s while `distiller.modeling` — concurrent, same pool, same model — hung. Every stall allowed to finish landed in a narrow band:

```

orchestrator.round_2 125.98s ok    spend_payments 129.50s ok
spend_payments       129.83s ok    crossbu        130.67s ok

```

while every failure died at **its own phase fence**, not at a natural duration:

```

report_agent      100.01s  <- report_agent_s was 100, now 200
distiller.modeling 120.01s  <- distiller_s = 120
general_specialist 25.00s   <- orch_plan_s = 25, round_1 ran 0.00s

```

So the fence a phase happens to carry decided whether it survived, while normal calls in those same turns took 2-13s. Widening fences only converts a failure into a 130s wait; escaping the stall is what recovers the time.

The multiplication trap

Agent-level and call-level retries compose. A specialist that retries once, each of whose calls stalls once, issues **up to 4 requests** — and its phase fence has to hold all of them. That is why the call-level budget is exactly one extra attempt rather than a general retry loop: the arithmetic has to stay inspectable against the fence above it.

Known inconsistencies

- **RESOLVED** — the "phase fence must be tighter than the per-call cap" invariant was backwards for **multi-round agents**. It holds for single-call phases (`screen`, `orch_plan`), where a tighter fence gives the clearer error. A specialist making four calls must be LOOSER than one call's cap or it can never complete more than one slow call. The rule is now split by phase kind, and `specialist_s/report_agent_s` are sized to hold one worst-case call ($40 + 140 = 180$) plus the agent's own rounds.
- **The reviewer shares `ORCH_PLAN_TIMEOUT_S` with the round-1 watchdog**, and they want opposite things: planning wants a tight 25s so a stall aborts and retries fast; the reviewer wants to outlast a stall. One knob cannot serve both — the reviewer is the agent that failed at exactly 25.00s.
- **The documented "slowest realistic path" does not fit the turn fence.** $30 + 25 + 240 + 25 + 150 = 470s$ against `turn_wall_clock_s = 360`. The guard test checks each phase against the fence individually, so nothing catches the sum.